

Assignment 2 – Command Line Arguments

Description:

This assignment is to write a C program that creates and manages a data struct, using pointers, memory buffers, and analyzing binary data in a hexadecimal dump.

Approach:

First, my approach was to read the assignment thoroughly. When I got to step one I noted that I should not overthink each step and just look at it as is for now, I read through the assignment probably around 3-4 times to first get a good idea of what the assignment is about and the first steps I needed to take. I also pulled up the class lecture/notes for reference, which is something I referred to often going forward in this assignment.

After thoroughly reading the header file, and following step three by properly naming my C file, implementing the proper code header, and noting key new concepts that we havent previously gone over such as, everytime I use a malloc I need to make sure to use a corresponding free or that there will be at least three arguments, but may there may be more, do not limit the number I was ready to start. I made a note because I believe keeping these things in mind will limit my chances of making slight errors.

To get started I first created a main function and called personalInfo info;. I then assigned some random values to all of the attributes in personalInfo and printed them out, I did this to make sure the program was up and running properly with the h file included and stdio.h included for access to personalInfo and printf.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought
forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that
all men are created equal."
First Name: Aidan
Last Name: Bayer-Calvert
Student ID: 922119403
Grade Level: 10
Languages Known (bitmap): 409
Message: Hello! This is Aidan's personal info.
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

Now I moved on to step 4, actually starting to code by using malloc and free to start working with the personalInfo struct. I first familiarized myself with malloc and how to actually use malloc to instansiate structs. I read through the malloc docs and watch some youtube videos to better understand this. In this process I also learned about the sizeof operator and how it basically tells you how many bytes are required to store the structure in memory. I learned that first I would need to use sizeof on the personalInfo struct to then be able to use malloc to allocate the exact amount of memory needed. This helped me allocate the correct amount of bytes into malloc. I then assigned the returned pointer from malloc to a type personalInfo *info variable, allowing me to access the fields. I then made sure to add free(info) at the end of my code to ensure after the program is finished the memory will be freed to avoid crashing. Next, I

moved on to assigning the first and last name. I used the argv array, which is an array of pointers, to access the first and second command-line arguments, corresponding to the first name and last name. These arguments are pointers to c strings, and I directly assigned them to info->firstName and info->lastName to populate the structure. I did a similar thing with info->studentID and info->level but in this case just hardcoded the values of my id(922119403) and enum grade level(SENIOR). Next I populated six coding languages into the languages field. At first I didnt know how to add multiple languages but I realized after doing some searching I could use | to help add multiple. Next I moved onto adding the message from the third argument into the info struct. I first noticed the 'hint' to keep in mind the character limit in the struct. So I wrote a loop to copy the third command-line argument character by character into the info->message field. I made sure to stop copying after 99 characters to respect the size limit of the message field. I was running into some issues with this, after some googling I realized I needed to use \0 not set the end of the string to NULL. After learning this I explicitly added logic to have the null terminator at the end of the message field or if the message was over 100 characters at the 100th character. Once I successfully completed this step, I printed the structure's fields to verify that all the values were correctly assigned and properly stored. Here is my successful output printed:

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation,
Liberty, and dedicated to the proposition that all men are created equal."
first name: aidan
last name: bayercalvert
student id: 922119403
grade: 10
languages: 266270
message: Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

Next, I moved on to step 5. I first looked at the .h file to find the writePersonalInfo function definition, then I called the function and saved the result in a variable called writePersonalInfoResult. In the github readme it mentioned that if calling the function has a return value of 0 it is a success so I added a check to verify this. Just a simple if else statement where if the result is a 0 it prints success message to the console and if the result is something other than a 0 it will print an error message to the console.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation,
Liberty, and dedicated to the proposition that all men are created equal."
success, write personal info succeeded.first name: aidan
last name: bayercalvert
student id: 922119403
grade: 10
languages: 266270
message: Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

After this, I moved on to step 6. I first started with just trying to iterate over the c string using getNext until the current char pointer was equal to NULL and printing all these results using "%s\n" on the nextString c string pointer. I accomplished this pretty quick, after I moved onto initializing the buffer. I had to do some research but learned to initialize a buffer it is just a char pointer using malloc and passing in the size. In this case I used the BLOCK_SIZE variable from the header file. Next I needed to figure out how to calculate how much bytes a string would take

up, I did this with some googling and found out that each char in a string is one byte so I could just find the length of the string excluding the null terminator to find the amount of bytes. I learned that C has a built in function to accomplish this called `strlen(str)`. I also quickly realized I would need a current buffer position variable that tracks where we are currently in the buffer. After initializing these, I created the calculations for figuring out if a string cannot fit in the remaining buffer space by doing current buffer position + string length, if this was greater than the block size I would commit the block and then set the current buffer position to 0. This was my edge case, if this didnt trigger then I would use `memcpy` to copy the string into the buffer and update the current buffer position. I also had some issues getting `memcpy` to work at the start. I was a little confused on the arguments and how to provide the destination, source, and num. After some googling I realized I needed to add `buffer + current buffer position` because `buffer` is a pointer to the start of the buffer and `current buffer position` would ensure the data is copied to the correct location. Then I could just use `nextString` as the source and the string length as the number of bytes or size. This seemed to work but one issue I quickly realized is that the buffer wouldnt always be completely full because if the string was too long then it would just commit the whole block and not be completely efficient. In order to resolve this I created a new variable called `stringPosition` and a while loop that continues processing the string until it has been exactly fully added to the buffer or in other words when the string position is less than the string length. This allowed me to break up the string into smaller parts and add them to the buffer in chunks, ensuring the buffer is always fully utilized before committing it. Within this new while loop, I calculated how much space was left in the buffer and determined how much of the string could fit into the remaining space. If the remaining portion of the string exceeded the buffer's available space, I only copied what would fit and committed the buffer afterward. This approach ensured that no part of the string was skipped and that the buffer was efficiently filled before committing. Finally, I added logic to handle any leftover data in the buffer after all strings were processed to make sure if there was no strings left and there was still data in the buffer it wouldnt accidentally be lost.

After this I moved on to step 7 where I called `checkIt()` and saved the result to a variable which I then returned at the end of `Main`. After this, I ran `make run` and it appeared to be working.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty,
and dedicated to the proposition that all men are created equal."
Success, write personal info succeeded..... CHECK .....
Running the check for aidan bayercalvert
Name check is 0 by 0
Student ID: 922119403, Grade Level: Senior
Languages: 266270
Message:
Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived
The Check Succeeded (0, 0)

END-OF-ASSIGNMENT
000000: 3E 53 65 D5 FF FF 00 00 44 53 65 D5 FF FF 00 00 | >Se???..DSe???..
000010: EB 6C F6 36 0A 00 00 00 1E 10 04 00 46 6F 75 72 | ?l?6.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 00 | tion, conceived.

student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ git add bayercalvert_aidan_HW2_main.c
```

My next step and final step was to complete step 8 or analyze the hexidump, I described this process in the analysis section.

Issues and Resolutions:

My first issue was trying to run my program after completing step 5. I noticed a lot of malloc errors as well as a lot of printf errors.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
bayercalvert_aidan_HW2_main.c: In function 'main':
bayercalvert_aidan_HW2_main.c:38:26: warning: implicit declaration of function 'malloc' [-Wimplicit-function-declaration]
   38 |     personalInfo *info = malloc(personalInfoByteSize);
      |                          ^
bayercalvert_aidan_HW2_main.c:27:1: note: include '<stdlib.h>' or provide a declaration of 'malloc'
   26 | #include <stdio.h>
     | ~~~~~
   27 | // include library that contains malloc operations
bayercalvert_aidan_HW2_main.c:38:26: warning: incompatible implicit declaration of built-in function 'malloc' [-Wbuiltin-declaration-mismatch]
   38 |     personalInfo *info = malloc(personalInfoByteSize);
      |                          ^
bayercalvert_aidan_HW2_main.c:38:26: note: include '<stdlib.h>' or provide a declaration of 'malloc'
bayercalvert_aidan_HW2_main.c:85:12: warning: character constant too long for its type
   85 |     printf('first name: ', info->firstName);
      |            ^~~~~~
bayercalvert_aidan_HW2_main.c:85:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conversion]
   85 |     printf('first name: ', info->firstName);
      |            ^~~~~~
      |            |
      |            int
In file included from bayercalvert_aidan_HW2_main.c:26:
/usr/include/stdio.h:356:43: note: expected 'const char * restrict' but argument is of type 'int'
   356 | extern int printf (const char *__restrict __format, ...);
      |
bayercalvert_aidan_HW2_main.c:86:12: warning: character constant too long for its type
   86 |     printf('last name: ', info->lastName);
      |            ^~~~~~
bayercalvert_aidan_HW2_main.c:86:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conversion]
   86 |     printf('last name: ', info->lastName);
      |            ^~~~~~
      |            |
      |            int
In file included from bayercalvert_aidan_HW2_main.c:26:
/usr/include/stdio.h:356:43: note: expected 'const char * restrict' but argument is of type 'int'
   356 | extern int printf (const char *__restrict __format, ...);
      |
bayercalvert_aidan_HW2_main.c:87:12: warning: character constant too long for its type
   87 |     printf('student id: ', info->studentID);
      |            ^~~~~~
bayercalvert_aidan_HW2_main.c:87:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conversion]
   87 |     printf('student id: ', info->studentID);
      |            ^~~~~~
      |            |
      |            int
In file included from bayercalvert_aidan_HW2_main.c:26:
/usr/include/stdio.h:356:43: note: expected 'const char * restrict' but argument is of type 'int'
   356 | extern int printf (const char *__restrict __format, ...);
      |
bayercalvert_aidan_HW2_main.c:88:12: warning: character constant too long for its type
   88 |     printf('grade: ', info->level);
      |            ^~~~~~
bayercalvert_aidan_HW2_main.c:88:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conversion]
   88 |     printf('grade: ', info->level);
      |            ^~~~~~
      |            |
      |            int
```

```
88 | printf('grade: ', info->level);
bayercalvert_aidan_HW2_main.c:88:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conve
88 | printf('grade: ', info->level);
|
|      |
|      int
In file included from bayercalvert_aidan_HW2_main.c:26:
/usr/include/stdio.h:356:43: note: expected 'const char * restrict' but argument is of type 'int'
356 | extern int printf (const char *__restrict __format, ...);
bayercalvert_aidan_HW2_main.c:89:12: warning: character constant too long for its type
89 | printf('languages: ', info->languages);
bayercalvert_aidan_HW2_main.c:89:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conve
89 | printf('languages: ', info->languages);
|
|      |
|      int
In file included from bayercalvert_aidan_HW2_main.c:26:
/usr/include/stdio.h:356:43: note: expected 'const char * restrict' but argument is of type 'int'
356 | extern int printf (const char *__restrict __format, ...);
bayercalvert_aidan_HW2_main.c:90:12: warning: character constant too long for its type
90 | printf('message: ', info->message);
bayercalvert_aidan_HW2_main.c:90:12: warning: passing argument 1 of 'printf' makes pointer from integer without a cast [-Wint-conve
90 | printf('message: ', info->message);
|
|      |
|      int
In file included from bayercalvert_aidan_HW2_main.c:26:
/usr/include/stdio.h:356:43: note: expected 'const char * restrict' but argument is of type 'int'
356 | extern int printf (const char *__restrict __format, ...);
bayercalvert_aidan_HW2_main.c:93:5: warning: implicit declaration of function 'free' [-Wimplicit-function-declaration]
93 | free(info);
bayercalvert_aidan_HW2_main.c:93:5: note: include '<stdlib.h>' or provide a declaration of 'free'
bayercalvert_aidan_HW2_main.c:93:5: warning: incompatible implicit declaration of built-in function 'free' [-Wbuiltin-declaration-m
bayercalvert_aidan_HW2_main.c:93:5: note: include '<stdlib.h>' or provide a declaration of 'free'
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new
Liberty, and dedicated to the proposition that all men are created equal."
Stack Trace: Error: signal 11::
./bayercalvert_aidan_HW2_main(printstack+0x2c) [0xb51381f2346c]
./bayercalvert_aidan_HW2_main(handler+0x40) [0xb51381f23574]
linux-vdso.so.1(__kernel_rt_sigreturn+0) [0xf710cde918f8]
/lib/aarch64-linux-gnu/libc.so.6(strchrnul+0xc) [0xf710cdd3370c]
/lib/aarch64-linux-gnu/libc.so.6(+0x609ec) [0xf710cdd009ec]
/lib/aarch64-linux-gnu/libc.so.6(_IO_printf+0x94) [0xf710cdcfc0a64]
./bayercalvert_aidan_HW2_main(main+0x130) [0xb51381f21b44]
/lib/aarch64-linux-gnu/libc.so.6(+0x273fc) [0xf710cdcc73fc]
/lib/aarch64-linux-gnu/libc.so.6(__libc_start_main+0x98) [0xf710cdcc74cc]
./bayercalvert_aidan_HW2_main(_start+0x30) [0xb51381f21930]
make: *** [Makefile:71: run] Error 11
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ =+
```

After looking through my code I quickly realized my include<stdlib.h> was somehow deleted. I made sure to add it and it fixed my malloc errors. I then moved on to the printf errors. I noticed that it was on all of my printf statements and my text inside of " were all highlighted. I did a quick google and realized you have to use double quotes in C. That fixed both the issues. The next error in that list was the 'too

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
bayercalvert_aidan_HW2_main.c: In function 'main':
bayercalvert_aidan_HW2_main.c:86:12: warning: too many arguments for format [-Wformat-extra-args]
  86 |     printf("first name: ", info->firstName);
      |     ^~~~~~
bayercalvert_aidan_HW2_main.c:87:12: warning: too many arguments for format [-Wformat-extra-args]
  87 |     printf("last name: ", info->lastName);
      |     ^~~~~~
bayercalvert_aidan_HW2_main.c:88:12: warning: too many arguments for format [-Wformat-extra-args]
  88 |     printf("student id: ", info->studentID);
      |     ^~~~~~
bayercalvert_aidan_HW2_main.c:89:12: warning: too many arguments for format [-Wformat-extra-args]
  89 |     printf("grade: ", info->level);
      |     ^~~~~~
bayercalvert_aidan_HW2_main.c:90:12: warning: too many arguments for format [-Wformat-extra-args]
  90 |     printf("languages: ", info->languages);
      |     ^~~~~~
bayercalvert_aidan_HW2_main.c:91:12: warning: too many arguments for format [-Wformat-extra-args]
  91 |     printf("message: ", info->message);
      |     ^~~~~~
Vmware-user | bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nat
Liberty, and dedicated to the proposition that all men are created equal."
first name: last name: student id: grade: languages: message: student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

After this, most of my errors cleared up but I still had some printf errors.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
bayercalvert_aidan_HW2_main.c: In function 'main':
bayercalvert_aidan_HW2_main.c:88:26: warning: format '%s' expects argument of type 'char *', but argument 2 has type 'int' [-Wformat=]
  88 |     printf("student id: %s\n", info->studentID);
      |                          ^~~~~~
      |                          |
      |                          char *      int
      |                          %d
bayercalvert_aidan_HW2_main.c:89:21: warning: format '%s' expects argument of type 'char *', but argument 2 has type 'unsigned int' [-Wformat=]
  89 |     printf("grade: %s\n", info->level);
      |                      ^~~~~~
      |                      |
      |                      char *      unsigned int
      |                      %d
bayercalvert_aidan_HW2_main.c:90:25: warning: format '%s' expects argument of type 'char *', but argument 2 has type 'int' [-Wformat=]
  90 |     printf("languages: %s\n", info->languages);
      |                          ^~~~~~
      |                          |
      |                          char *      int
      |                          %d
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nat
Liberty, and dedicated to the proposition that all men are created equal."
first name: aidan
last name: bayercalvert
Stack Trace: Error: signal 11::
./bayercalvert_aidan_HW2_main(printstack+0x2c) [0xc38f8876346c]
./bayercalvert_aidan_HW2_main(handler+0x40) [0xc38f88763574]
linux-vdso.so.1(__kernel_rt_sigreturn+0) [0xeaadd61ea8f8]
/lib/aarch64-linux-gnu/libc.so.6(+0x998cc) [0xeaadd60898cc]
/lib/aarch64-linux-gnu/libc.so.6(+0x61f68) [0xeaadd6051f68]
/lib/aarch64-linux-gnu/libc.so.6(_IO_printf+0x94) [0xeaadd6040a64]
./bayercalvert_aidan_HW2_main(main+0x160) [0xc38f88761b74]
/lib/aarch64-linux-gnu/libc.so.6(+0x273fc) [0xeaadd60173fc]
/lib/aarch64-linux-gnu/libc.so.6(__libc_start_main+0x98) [0xeaadd60174cc]
./bayercalvert_aidan_HW2_main(_start+0x30) [0xc38f88761930]
make: *** [Makefile:71: run] Error 11
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

I quickly realized it had to do with format specifiers, I added %s\n to every print statement. I then encountered another error and realized that %s doesnt work with every statement and that I also needed %d.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation,
conceived in Liberty, and dedicated to the proposition that all men are created equal."
first name: aidan
last name: bayercalvert
student id: 922119403
grade: 10
languages: 266270
message: Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

This fixed everything and my output for step four now seemed correct.

The next issue I had was trying to run my program after creating my finalized version of step 6.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
bayercalvert_aidan_HW2_main.c: In function 'main':
bayercalvert_aidan_HW2_main.c:111:24: warning: initialization discards 'const' qualifier from pointer target type [-Wdiscarded-qualifiers]
111 |     char *nextString = getNext();
    |                        ^
bayercalvert_aidan_HW2_main.c:120:27: warning: implicit declaration of function 'strl'; did you mean 'strtol'? [-Wimplicit-function-declaration]
120 |     int stringBytes = strl(nextString);
    |                       ^
    |                       strtol
bayercalvert_aidan_HW2_main.c:145:13: warning: implicit declaration of function 'memcpy' [-Wimplicit-function-declaration]
145 |     memcpy(buffer + currBufferPosition, nextString + stringPosition, copyLength);
    |     ^
bayercalvert_aidan_HW2_main.c:29:1: note: include '<string.h>' or provide a declaration of 'memcpy'
28 | #include <stdlib.h>
+++ |+#include <string.h>
29 |
bayercalvert_aidan_HW2_main.c:145:13: warning: incompatible implicit declaration of built-in function 'memcpy' [-Wbuiltin-declaration-mismatch]
145 |     memcpy(buffer + currBufferPosition, nextString + stringPosition, copyLength);
    |     ^
bayercalvert_aidan_HW2_main.c:145:13: note: include '<string.h>' or provide a declaration of 'memcpy'
bayercalvert_aidan_HW2_main.c:164:20: warning: assignment discards 'const' qualifier from pointer target type [-Wdiscarded-qualifiers]
164 |     nextString = getNext();
    |                ^
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
/usr/bin/ld: bayercalvert_aidan_HW2_main.o: in function 'main':
/home/student/Desktop/csc415-assignment-2-bufferandstruct-abccodes/bayercalvert_aidan_HW2_main.c:120: undefined reference to `strl'
collect2: error: ld returned 1 exit status
make: *** [Makefile:65: bayercalvert_aidan_HW2_main] Error 1
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ ^C
```

The first error message after googling appears to be how I defined my getNext() function to a char*nextString variable. Apparently getNext returns a const char* type variable because it is immutable data, I updated this and it appeared to fix the issue. The next issue encountered was implicit declaration of function strl. This was from a typo, I needed to use strlen not strl. The next error was from my memcpy function call. I realized quickly that it was because I didnt properly import the <string.h> header.

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic -I.
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal."
Success, write personal info succeeded.----- CHECK -----
Running the check for aidan bayercalvert
Name check is 0 by 0
Student ID: 922119403, Grade Level: Senior
Languages: 266270
Message:
Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal."

The Check Succeeded (0, 0)

END-OF-ASSIGNMENT
000000: 3E 33 8D E6 FF FF 00 00 44 33 8D E6 FF FF 00 00 | >3????..D3????..
000010: EB 6C F6 36 0A 00 00 00 1E 10 04 00 46 6F 75 72 | ?l?6.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 00 | tion, conceived.

student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

This appeared to fix all my issues.

Analysis:

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal."
Success, write personal info succeeded.----- CHECK -----
Running the check for aidan bayercalvert
Name check is 0 by 0
Student ID: 922119403, Grade Level: Senior
Languages: 266270
Message:
Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal."

The Check Succeeded (0, 0)

END-OF-ASSIGNMENT
000000: 3E 43 99 C6 FF FF 00 00 44 43 99 C6 FF FF 00 00 | >C????..DC????..
000010: EB 6C F6 36 0A 00 00 00 1E 10 04 00 46 6F 75 72 | ?l?6.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 00 | tion, conceived.

student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

```
000000: 3E 43 99 C6 FF FF 00 00 44 43 99 C6 FF FF 00 00 | >C????..DC????..
000010: EB 6C F6 36 0A 00 00 00 1E 10 04 00 46 6F 75 72 | ?l?6.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E | score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 | years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 | athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E | orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 | tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 00 | tion, conceived.
```

(zoomed in image with annotations)

To get started, I first read all of step 8. When looking at the output of `checkIt()` in the console, I was at first very confused and didn't know where to start. After doing some searching I realized that I needed to look at my `personallInfo` struct as that was what was being displayed in the hexadump. I figured out by reading step 8 that all a hexadump is, is displaying the binary data which can be referenced to the `personallInfo` struct by comparing the bytes to its hexadecimal value converting to decimal. To see if I could just test this out on a small scale I first just started with trying with the first variable in the `personallInfo` struct, `firstName`. I did a google search to verify that pointers on 64-bit systems are 8 bytes in size meaning a pointer like `char *firstName` occupies 8 bytes or 64 bits. At first I was running into a lot of trouble because I was trying to convert the `firstName` to hexadecimal values to then verify in the hexidump. After some searching, I realized the hexidump is just storing the pointer so the actual value of first name would not be shown. Because `firstName` is 8 bytes, it occupies the values of `3E 43 99 C6 FF FF 00 00` or `0x0000, 0x0007`. At first I was also really confused by the `00 00`, but apparently for pointers it is common that there are `00`s because they could be zeroed out higher bits. I also learned that each `XX` value in the hexidump was basically equivalent to one byte meaning these values matched the 8 byte pointer. This is shown in the orange highlighted section of the hexidump.

Moving onto `lastName`, it is also an 8 byte pointer, so it would represent the values of `44 43 99 C6 FF FF 00 00` or `0x0008, 0x000F`. This is shown in the red highlighted section of the hexidump.

Moving on to student ID, I googled that an integer in a 64 bit system occupies 4 bytes in space. So student id, `922119403`, occupies `EB 6C F6 36` or `0x0010 to 0x0013`. Unlike the pointers for `firstName` and `lastName`, student id is just an integer so it can be converted to a hexadecimal value. The conversion for `922119403` to hexadecimal is `0x36F66CEB`. When trying to compare at first it didn't seem like they matched up but I realized they actually did but in reverse order. so reversing `0x36F66CEB` and reversing all the characters is equal to `EB6CF636` which is equivalent to `EB 6C F6 36`. I did some searching and saw this was called little-endian format, I'm not familiar with this at all but it seems to match up so I am fairly confident this is correct. This is shown in the blue highlighted section.

Moving onto the next variable, the enum grade level, I looked it up and enums appear to also be 4 bytes. So in the hexidump it would occupy values of `0x0014 to 0x0017` or `0A 00 00 00`. In the enum declaration inside of the `personallInfo` struct it says `FRESHMAN=7` meaning the attribute I inputted, `SENIOR`, would contain a value of 10. When converting 10 to hexadecimal it is equivalent to `0x0A` which matches with `0A 00 00 00`. Since 10 only contains 1 byte only 1 out of 4 bytes are being used which matches up. This is shown in the purple highlighted section.

The next variable is `languages`, which is a `int` declaration referencing bitmap values for different coding languages which are defined in the header file. First I included 6 different languages, `JAVA`, `JAVASCRIPT`, `SWIFT`, `PYTHON`, `MIPS`, and `C++`. The corresponding hexadecimal values of the selected languages are `0x00000002`, `0x00000004`, `0x00000008`, `0x00001000`, `0x00040000`, and `0x00000010`. After doing some research I learned that you can add up these values because that is what eventually gets stored in hexadecimal. The sum of all these up equals `0x0004101E`. When using little-endian format again it would be equivalent to `1E 10 04 00`. When looking at `0x0018`

to 0x001B it holds the values 1E 10 04 00 which matches, verifying the accuracy of the hexidump result. This is shown in the neon green highlighted section.

The next personalInfo attribute is char message[100] which represents a fixed size array of 100. Each character represents one byte so in total it is equivalent to 100 bytes. When looking at the input it is: "Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal." This message in total is 176 characters long meaning not all 100 characters will fit because each character equals a byte meaning 176 bytes cannot fit in 100 bytes. To verify all the characters that are stored in this messages characters array, we can look at the first 100 characters which represent: "Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived"(99 characters) plus the null terminator(\0). I googled that we can verify this by converting this whole message to ascii then looking at the hexidump values. So after plugging the 100 character message into a ascii calculator I got the result of: "46 6F 75 72 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64", and figured out that there was no need for endian format because ascii characters are already represented as single bytes. By starting at 0x001C every single value matches up until 0x007F where 0x007F is 00 which is expected because 0x007F would hold the null terminator. This means that all 99 characters plus the null terminator all correlate and match up, proving the hexidump is correctly matching the data in personalInfo. This is shown in the pink highlighted sections, the maroon highlighted 00 represents the null terminator.

Screen shot of compilation:

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make clean
rm bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make
gcc -c -o bayercalvert_aidan_HW2_main.o bayercalvert_aidan_HW2_main.c -g -rdynamic -I.
gcc -o bayercalvert_aidan_HW2_main bayercalvert_aidan_HW2_main.o assignment2M1.o -g -rdynamic
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```

Screen shot(s) of the execution of the program:

(next page)

```
student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$ make run
./bayercalvert_aidan_HW2_main aidan bayercalvert "Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liber
ty, and dedicated to the proposition that all men are created equal."
Success, write personal info succeeded.----- CHECK -----
Running the check for aidan bayercalvert
Name check is 0 by 0
Student ID: 922119403, Grade Level: Senior
Languages: 266270
Message:
Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived

The Check Succeeded (0, 0)

END-OF-ASSIGNMENT
000000: 3E A3 9D CB FF FF 00 00 44 A3 9D CB FF FF 00 00 | >?????.D?????..
000010: EB 6C F6 36 0A 00 00 00 1E 10 04 00 46 6F 75 72 | ?l?6.....Four
000020: 20 73 63 6F 72 65 20 61 6E 64 20 73 65 76 65 6E |  score and seven
000030: 20 79 65 61 72 73 20 61 67 6F 20 6F 75 72 20 66 |  years ago our f
000040: 61 74 68 65 72 73 20 62 72 6F 75 67 68 74 20 66 |  athers brought f
000050: 6F 72 74 68 20 6F 6E 20 74 68 69 73 20 63 6F 6E |  orth on this con
000060: 74 69 6E 65 6E 74 2C 20 61 20 6E 65 77 20 6E 61 |  tinent, a new na
000070: 74 69 6F 6E 2C 20 63 6F 6E 63 65 69 76 65 64 00 |  tion, conceived.

student@student:~/Desktop/csc415-assignment-2-bufferandstruct-abccodes$
```